

The Pentagon, Machine-Checked

Abel's five-term relation for the dilogarithm in Lean 4

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June 2026

Abstract

I give a machine-checked, **sorry-free** proof, in Lean 4 over Mathlib, of *Abel's five-term relation* for the dilogarithm. With L the Rogers L -function, it reads, for $x, y \in (0, 1)$,

$$L(x) + L(y) = L(xy) + L\left(\frac{x(1-y)}{1-xy}\right) + L\left(\frac{y(1-x)}{1-xy}\right).$$

This single identity is the whole weight-2 theory in one line: by theorems of Wojtkowiak and de Jeu, *every* functional equation of Li_2 with rational arguments is a consequence of it, and it is the defining relation of the Bloch group and the pentagon of cluster algebras. The proof is the classical differentiation argument made fully rigorous: fixing y , the five-term difference has identically zero derivative on $(0, 1)$ – nine logarithmic terms collapse into a basis of five and the rest is **ring** – so it is constant, and its boundary value as $x \rightarrow 0^+$ is computed by squeezing $\log x \cdot \log(1-x) \rightarrow 0$. The development is self-contained: Mathlib carries no dilogarithm, so this sits on a from-scratch Lean construction of Li_2 . I am exact about the boundary: the mathematics is entirely classical (Spence, Abel, Rogers; Wojtkowiak's and de Jeu's universality); the contribution is the formalization – to the best of my knowledge the first machine-checked proof of the five-term relation, and the first machine-checked dilogarithm, in any proof assistant – together with three reusable building blocks. The headline theorem and its blocks depend only on `propxet`, `Classical.choice` and `Quot.sound`.

1 One identity to rule them all

Put $x = y = \frac{1}{2}$. The five-term relation collapses, after $\frac{(1/2)(1/2)}{1-1/4} = \frac{1}{3}$, to

$$2 L\left(\frac{1}{2}\right) = L\left(\frac{1}{4}\right) + 2 L\left(\frac{1}{3}\right), \quad \text{i.e.} \quad L\left(\frac{1}{4}\right) + 2 L\left(\frac{1}{3}\right) = \frac{\pi^2}{6},$$

since $L(\frac{1}{2}) = \pi^2/12$. That a sum of three dilogarithm values at $\frac{1}{4}$ and $\frac{1}{3}$ should be exactly $\pi^2/6$ is not obvious; it is one of infinitely many shadows of a single two-variable law. The dilogarithm $\text{Li}_2(x) = \sum_{n \geq 1} x^n/n^2$ obeys a handful of functional equations. Two are elementary – Euler's *reflection* $\text{Li}_2(x) + \text{Li}_2(1-x) = \frac{\pi^2}{6} - \log x \log(1-x)$ and *Landen's transformation* – and one is not: the *five-term relation*, found by Spence (1809) and Abel (1830) and rediscovered ever since. It is the deepest weight-2 identity, and the only one that truly needs two variables.

*Formalized with AI assistance (Claude, Anthropic); the mathematics and all claims are the author's responsibility. The exact boundary of the contribution is stated in the abstract and Section 4.

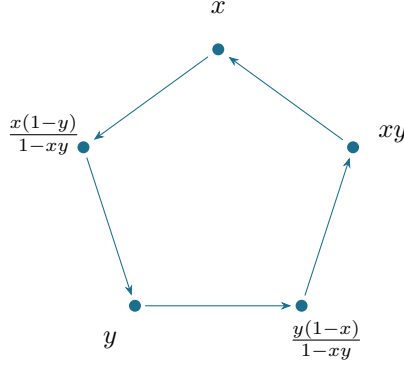


Figure 1: The five arguments on a cycle. The five-fold symmetry of this pentagon is the combinatorial shadow of the Bloch-group relation and of the cluster-algebra mutation pentagon.

Its centrality is a theorem, not a slogan. Wojtkowiak proved that every one-variable functional equation of Li_2 with rational arguments is a $\mathbb{Z}$ -linear combination of instances of the five-term relation; de Jeu extended this to any number of variables [1, 2]. The same identity *defines* the Bloch group, so it governs scissors-congruence invariants and the volumes of hyperbolic 3-manifolds [3]; in cluster algebras it is the *pentagon*, the relation of five mutations [4]. Write the five arguments on a cycle (Figure 1) and the five-fold symmetry that powers all of this appears.

The cleanest statement uses the *Rogers L-function* $L(x) := \text{Li}_2(x) + \frac{1}{2} \log x \log(1-x)$, which absorbs the stray logarithms of the raw Li_2 form and turns the relation into a pure sum of five L values:

$$L(x) + L(y) = L(xy) + L\left(\frac{x(1-y)}{1-xy}\right) + L\left(\frac{y(1-x)}{1-xy}\right), \quad x, y \in (0, 1).$$

The state of formalization. Mathlib, the largest formal library, carries no dilogarithm: neither Li_2 nor any of its functional equations. A check of the Lean, Isabelle and Rocq (Coq) libraries in June 2026 found no formalization of the five-term relation, nor of Li_2 as an analytic object beyond elementary special values. What follows therefore sits on a from-scratch Lean construction of Li_2 – its series, derivative $-\log(1-x)/x$, reflection, Landen, duplication – of which the five-term relation is the capstone. With it, the two-variable identities and the polylogarithm *ladders* (the golden-ratio and quartic-base evaluations) become corollaries rather than separate formalization targets.

2 The statement, and the idea of the proof

Theorem 1 (Dilog.rogersL_fiveterm). *For all $x, y \in (0, 1)$,*

$$L(x) + L(y) = L(xy) + L\left(\frac{x(1-y)}{1-xy}\right) + L\left(\frac{y(1-x)}{1-xy}\right).$$

I prove it the textbook way – by differentiation – but with every step made rigorous. Fix $y \in (0, 1)$ and set

$$G(t) = L(t) - L(ty) - L\left(\frac{t(1-y)}{1-ty}\right) - L\left(\frac{y(1-t)}{1-ty}\right).$$

The theorem is $G(x) = -L(y)$. Three movements.

1. The derivative vanishes. For $x, y \in (0, 1)$ all three composite arguments again lie in $(0, 1)$, so every term is differentiable. With $L'(t) = -\frac{1}{2}(\frac{\log(1-t)}{t} + \frac{\log t}{1-t})$, the chain rule writes $G'(t)$ as four such expressions weighted by the derivatives of the composite arguments. Every logarithm that occurs – of t , of ty , of the two fractional arguments and their complements – expands, by $\log(ab) = \log a + \log b$ and $\log(a/b) = \log a - \log b$, into the five-element basis $\{\log t, \log y, \log(1-t), \log(1-y), \log(1-ty)\}$. In that basis $G'(t)$ is a rational expression whose every coefficient cancels: $G' \equiv 0$. In Lean this is the one delicate step (`fiveterm_hasDerivAt_zero`): the nine logarithms are rewritten into the basis and the rational remainder is cleared and closed by `ring`. (The cancellation was checked numerically to thirty digits before it was formalized – and the formalization caught a subtlety the eye misses: `field_simp` silently reorders $x \cdot y$ to $y \cdot x$ and then needs $1 - yx \neq 0$ in hand to discharge the resulting inverse, a fact that simply had to be in scope at the right moment.)

2. Hence G is constant. A function with vanishing derivative on the open connected interval $(0, 1)$ is constant there – Mathlib’s `IsOpen.is_const_of_deriv_eq_zero`, the mean-value theorem in disguise. So $G(x) = G(t)$ for every $t \in (0, 1)$.

3. The boundary value. As $t \rightarrow 0^+$, $ty \rightarrow 0$, $\frac{t(1-y)}{1-ty} \rightarrow 0$ and $\frac{y(1-t)}{1-ty} \rightarrow y$, so $G(t) \rightarrow L(0) - L(0) - L(0) - L(y) = -L(y)$ – provided L is right-continuous at 0 with $L(0) = 0$. This is the one analytic subtlety. $L(t) = \text{Li}_2(t) + \frac{1}{2} \log t \log(1-t)$, and while Li_2 is continuous, $\log t \cdot \log(1-t)$ is an indeterminate $(-\infty) \cdot 0$. It is tamed by a squeeze: on $(0, \frac{1}{2}]$ one has $-\log(1-t) \leq 2t$ (from $\log z \geq 1 - z^{-1}$), so

$$0 \leq \log t \cdot \log(1-t) = (-\log t)(-\log(1-t)) \leq 2t(-\log t) = 2(-t \log t) \rightarrow 0,$$

because $t \log t \rightarrow 0$ – Mathlib’s `Real.continuous_mul_log`. Thus $G \rightarrow -L(y)$; being constant and equal to $G(x)$ near 0, it *equals* $-L(y)$, which is Theorem 1. $\square$

The recipe: read off a π^2 -evaluation in three lines

Pick any $x, y \in (0, 1)$ at which the four arguments are nice. (1) Form xy , $\frac{x(1-y)}{1-xy}$, $\frac{y(1-x)}{1-xy}$. (2) Apply $L(x) + L(y) =$ their sum, with a machine-checked guarantee. (3) Substitute any L values you know.

Worked: take $x = y = \frac{1}{2}$. Then $xy = \frac{1}{4}$ and both fractions are $\frac{1}{3}$, so $2 L(\frac{1}{2}) = L(\frac{1}{4}) + 2 L(\frac{1}{3})$. Since $L(\frac{1}{2}) = \text{Li}_2(\frac{1}{2}) + \frac{1}{2} \log^2 2$ and $\text{Li}_2(\frac{1}{2}) = \frac{\pi^2}{12} - \frac{1}{2} \log^2 2$, the log terms cancel and $L(\frac{1}{2}) = \frac{\pi^2}{12}$; hence $L(\frac{1}{4}) + 2 L(\frac{1}{3}) = \frac{\pi^2}{6}$ – a closed form for three dilogarithm values, certified.

3 The building blocks

The headline rests on three reusable pieces, each `sorry`-free with the standard three axioms.

Lemma 1 (`Dilog.hasDerivAt_rogersL`). On $(0, 1)$, $L'(x) = -\frac{1}{2}(\frac{\log(1-x)}{x} + \frac{\log x}{1-x})$.

The same derivative scheme already used for Euler reflection and Landen: combine $\text{Li}_2'(x) = -\log(1-x)/x$ with the product rule for $\frac{1}{2} \log x \log(1-x)$.

Lemma 2 (`Dilog.tendsto_rogersL_nhdsGT_zero`). $L(t) \rightarrow 0$ as $t \rightarrow 0^+$.

The Li_2 part is continuous on $[-1, 1]$; the log-product is squeezed to 0 by $2(-t \log t)$. This is the only place the proof leaves pure algebra for analysis.

Lemma 3 (`Dilog.continuousAt_rogersL`). L is continuous at every $x \in (0, 1)$.

Needed for the one surviving boundary term, $L\left(\frac{y(1-t)}{1-ty}\right) \rightarrow L(y)$.

4 The boundary of the contribution

Everything here that is mathematics is classical. The five-term relation is Spence’s and Abel’s; the Rogers L -function is Rogers’s (1907); the universality statements are Wojtkowiak’s and de Jeu’s; the differentiation proof is standard. I claim no new identity and no new inequality. The contributions are three: the *formalization* (to my knowledge the first machine-checked five-term relation, on the first machine-checked dilogarithm, in any proof assistant); the *reusable infrastructure* – the Rogers- L derivative, the right-continuity at 0 via the $-t \log t$ squeeze, and the interior continuity – which is exactly what any further weight-2 development (ladders, the Bloch group, the Bloch–Wigner five-term) will need; and the *proof-engineering record*, where the nine-into-five logarithmic cancellation is the kind of step a formal check earns its keep on.

On methodology. The formalization was done with an AI assistant (Claude, Anthropic), with the central cancellation verified numerically to thirty digits before formalizing and every named theorem checked by `#print axioms` to depend only on `propext`, `Classical.choice` and `Quot.sound`. Correctness does not rest on the assistant: the Lean kernel checks every proof. The mathematics and the claims are the author’s responsibility.

5 Open roads

1. **Ladders as corollaries.** With the five-term relation, the golden-ratio ladders $L(1/\varphi) = \pi^2/10$, $L(1/\varphi^2) = \pi^2/15$ and their quartic-base relatives are linear-algebra consequences rather than separate proofs; mechanizing this in Lean is the natural next step.
2. **The Bloch–Wigner five-term.** The same relation holds for the single-valued Bloch–Wigner function on $\mathbb{C}$, where it underlies the hyperbolic-volume and scissors-congruence applications; formalizing it is the complex-analytic sequel.
3. **Universality.** Wojtkowiak’s reduction of every rational one-variable equation to the five-term would be a deep formalization target, of which this is the indispensable first brick.

Reproducing

```
git clone https://github.com/karlesmarin/godsil-gutman-lean
cd godsil-gutman-lean && lake exe cache get
lake build Dilog.FiveTerm
```

Lean v4.30.0-rc2, Mathlib pin 701fb6e9. The relation is `Dilog.rogersL_fiveterm` in `Dilog/FiveTerm.lean`; its supports are `hasDerivAt_rogersL`, `fiveterm_hasDerivAt_zero`, `tendsto_rogersL_nhdsGT_zero` and `continuousAt_rogersL`. Each is checked by `#print axioms` to depend only on `propext`, `Classical.choice`, `Quot.sound`.

References

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